CLIL UNIT 2 • SENSES, NERVES & WILDLIFE HEROES

# INTERPRETING OUR WORLD & ENDANGERED ANIMALS!

The 5 Senses, Nervous System, Reflexes, Animal Superpowers & Comparatives

*(Fusión didáctica: Los 5 sentidos, sistema nervioso, reflejos, fauna amenazada y comparativos)*

**1. SENSES**
5 Receptors

**2. NERVOUS**
Brain & Reflexes

**3. WILDLIFE**
Endangered Species

**4. GRAMMAR**
Comparatives



# 1. The Interaction Function

La función de relación: percibir estímulos, procesar información y actuar

## 📡 1. Receptor

### Detect Stimuli

*Captar estímulos del entorno*

Sense organs contain specialized sensory receptor cells that detect light, sound waves, chemicals, heat, or touch pressure.

👉 ***Eyes spot a falling boulder!***

## 🧠 2. Coordinator

### Process & Decide

*Procesar y ordenar respuesta*

Sensory nerves send electric pulses to the Central Nervous System (Brain and Spinal Cord) to decode the signal and choose an action.

👉 ***Brain commands: "Jump to safety!"***

## 🏃 3. Effector

### Execute Response

*Ejecutar la respuesta motora*

Motor nerves send rapid electrical triggers to muscles (locomotor response) or glands to produce motion or adrenaline.

👉 ***Leg muscles contract instantly!***

# 2. The 5 Senses & Specialized Receptors

Los cinco órganos de los sentidos y el tipo de receptor celular que poseen

## 👁️ Sight

**Eyes**

**PHOTORECEPTORS**

Retina cells (rods and cones) capture light and colours. Optic nerve to brain.

## 👂 Hearing

**Ears**

**MECHANORECEPTORS**

Cochlea hair cells vibrate with sound waves and maintain physical balance.

## 👃 Smell

**Nose**

**CHEMORECEPTORS**

Olfactory cells detect gas molecules in the air. Olfactory nerve to brain.

## 👅 Taste

**Tongue**

**CHEMORECEPTORS**

Taste buds identify sweet, salty, sour, bitter, and umami tastes in liquids.

## 🖐️ Touch

**Skin**

**THERMO & MECHANO**

Nerve endings in dermis sense temperature, pressure, textures, and pain.

# 3. The Nervous System

Sistema Nervioso Central (SNC) y Sistema Nervioso Periférico (SNP)

## 🧠 Central Nervous System (CNS)

*Sistema Nervioso Central (Encéfalo y Médula)*

- **Cerebrum (Cerebro):** Controls thought, memory, voluntary motion, emotions, and sensory perception.
- **Cerebellum (Cerebelo):** Coordinates precision movement, muscle balance, and athletic agility.
- **Brainstem (Bulbo raquídeo):** Manages involuntary vital life processes (heartbeat, digestion, breathing).
- **Spinal Cord (Médula espinal):** Highway of nerve signals connecting the brain to body; manages quick reflexes!

## ⚡ Peripheral Nervous System (PNS)

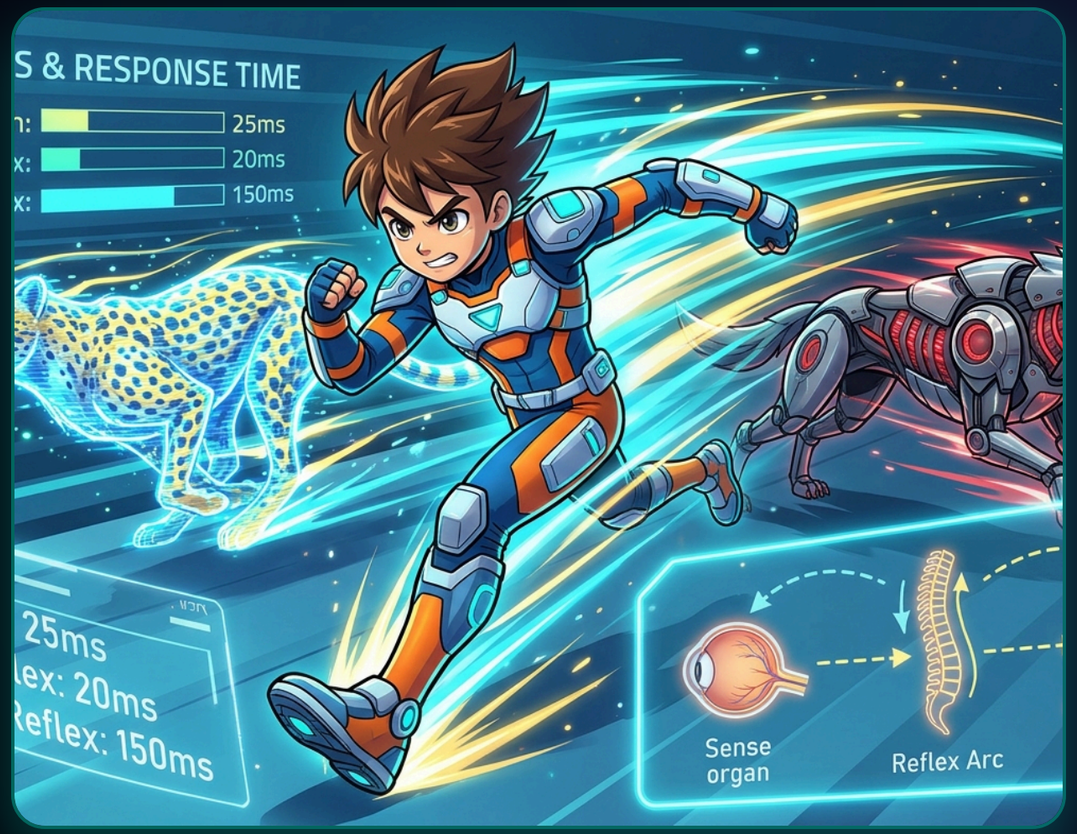
*Sistema Nervioso Periférico (Red de nervios)*

A massive network of billions of nerves running across every millimeter of the human body:

1. **Sensory Nerves (Nervios sensitivos):** Carry signals FROM the sense organs INWARD to the brain.
2. **Motor Nerves (Nervios motores):** Carry commands FROM the brain OUTWARD to the muscles and glands.

# 4. Voluntary Actions vs. Reflexes!

Actos voluntarios conscientes frente a arcos reflejos involuntarios de supervivencia




## 🎯 Voluntary Action (Acto Voluntario)

**Conscious decision:** The stimulus reaches the brain, you consciously think and decide, then send motor commands to your limbs.

👉 *Example: Deciding to pass a basketball or kick a football. Takes ~200 milliseconds.*

## Reflex Action (Acto Reflejo)

**Instant survival loop:** The signal bypasses the brain! Sensory nerves go straight to the **spinal cord**, which immediately fires a motor command before you even feel pain.

👉 *Example: Pulling your hand off a red-hot stove. Takes ~20 milliseconds!*

🗣️ **English Grammar Link:** *"Reflex actions are **faster than** voluntary actions because they travel a **shorter** neural path."*

# 5. Endangered Animals • Especies Amenazadas

Animales salvajes en peligro de extinción, sus hábitats y las causas de su declive

### Iberian Lynx
LINCE IBÉRICO

The most endangered wild feline in Europe. Lives in Mediterranean scrubland (Extremadura and Andalusia).

### Blue Whale
BALLENA AZUL

The largest animal on planet Earth. Threatened by ocean shipping traffic, plastic pollution, and whaling.

### Polar Bear
OSO POLAR

Apex Arctic predator hunting on sea ice. Endangered due to global warming and melting polar ice caps.

### Mountain Gorilla
GORILA DE MONTAÑA

Gentle giant primate living in high volcanic cloud forests. Threatened by deforestation and illegal poaching.

### Imperial Eagle
ÁGUILA IMPERIAL

Magnificent raptor with acute eyesight. Protected in Spanish National Parks like Monfragüe (Extremadura).

### Iberian Wolf
LOBO IBÉRICO

Keystone predator with incredible olfactory smell and stamina. Crucial for balancing forest ecosystems.

# 6. Animal Super Senses vs. Humans

Comparativa biológica: las adaptaciones sensoriales de la fauna salvaje frente a los sentidos humanos

## 🦅 Eagle Vision

### Super Sight

An imperial eagle's retina has **5 times more photoreceptors** than a human eye!

**Comparative:**
*"An eagle's eyesight is **sharper than** a human's sight. An eagle can spot a rabbit from 3 kilometers away!"*

## 🐾 Lynx Hearing

### Super Ears

The Iberian lynx has tufted ears acting as antennas that swivel 180 degrees to detect high-frequency vibrations.

**Comparative:**
*"A lynx hears **better than** a human. Its ears are **more sensitive than** our ears."*

## 🐺 Wolf Smell

### Super Olfaction

A wolf possesses **300 million olfactory receptors** (humans only have 5 million).

**Comparative:**
*"A wolf's sense of smell is **60 times stronger than** our sense of smell!"*

# 7. Grammar: Comparative Adjectives

Reglas para comparar animales, sentidos, velocidad y tamaños en inglés

## 1. Short Adjectives (+er than)

*Adjetivos cortos de 1 o 2 sílabas*

- fast $ ightarrow$ **faster than**
- big $ ightarrow$ **bigger than** (dobla consonante)
- heavy $ ightarrow$ **heavier than** (-y pasa a -ier)
- wild $ ightarrow$ **wilder than**

👉 *"A cheetah is **faster than** a human athlete."*

## 2. Long Adjectives (more ... than)

*Adjetivos largos de 2 o más sílabas*

- endangered $ ightarrow$ **more endangered than**
- intelligent $ ightarrow$ **more intelligent than**
- dangerous $ ightarrow$ **more dangerous than**
- sensitive $ ightarrow$ **more sensitive than**

👉 *"The lynx is **more endangered than** the wolf."*

## 3. Irregulars & Equality

*Comparativos irregulares y de igualdad*

- good $ ightarrow$ **better than**
- bad $ ightarrow$ **worse than**
- far $ ightarrow$ **farther / further than**
- **as + adj + as:** *as fast as* (tan rápido como)

👉 *"An owl sees **better than** humans at night."*

# 8. Reaction Time: Animals vs. Athletes

La velocidad de reacción en milisegundos: el tiempo que tarda un estímulo en ejecutarse

## 🦅 Eagle

**10 ms**

ULTRA-REFLEX

Detects prey motion in mid-air and adjusts wing angle in a split second.

## 🐾 Iberian Lynx

**25 ms**

AMBUSH SPEED

Jumps instantly when hearing a rabbit twitching grass 15 meters away.

## 🏃 Olympic Sprinter

**150 ms**

HUMAN PEAK

Trained auditory response to the sound of the starting gun.

## 👱 Average Human

**250 ms**

NORMAL REFLEX

Time needed for eye photoreceptors to signal brain and push a button.

# 9. Why Are Wild Animals Endangered?

Las 3 grandes amenazas ecológicas y nuestro deber ético de proteger la biodiversidad

## 🚜 1. Habitat Loss

**Destruction of Habitats**

*Pérdida y fragmentación de hábitats*

Deforestation, expanding motorways, and urbanization cut ecosystems into small fragments, leaving wild animals without hunting grounds or mates.

## 🌡️ 2. Climate Change

**Global Warming**

*Cambio climático y calentamiento global*

Rising ocean temperatures bleach coral reefs, melt Arctic sea ice where polar bears hunt, and cause severe droughts that dry up wetlands.

## 🛡️ 3. Human Solutions!

**National Parks & Laws**

*Espacios protegidos y cría en cautividad*

Thanks to breeding programs in Extremadura and strict environmental laws, the Iberian lynx population has grown from fewer than 100 to over 1,500!

# 10. Healthy Habits for Brain & Senses

Pautas médicas para proteger los receptores sensoriales y el sistema nervioso

## 😴 9-10h Sleep

**Deep Sleep**

DESCANSO CEREBRAL

During sleep, your brain cleans out toxic waste, recharges neurotransmitters, and consolidates memory.

## 🎧 Safe Volume

**Ear Protection**

CUIDADO AUDITIVO

Loud music over 85 decibels destroys delicate cochlea hair cells permanently. Keep headphones at 60% volume!

## 📱 Screen Limits

**Eye Hygiene**

DESCANSO VISUAL

Follow the 20-20-20 rule: Every 20 minutes looking at screens, look 20 feet (6m) away for 20 seconds.

## 🥦 Brain Foods

**Omega-3 & Water**

NUTRICIÓN NEURONAL

Nuts, fish, fruits, and hydration keep the myelin sheath around nerves strong and fast.

# 11. Vocabulary Vault • Banco Bilingüe

16 términos clave de Ciencias e Inglés con su traducción en castellano

**Stimulus**
ESTÍMULO
Change detected by sense organs.

**Receptor**
RECEPTOR SENSORIAL
Specialized cell capturing signals.

**Brain**
CEREBRO / ENCÉFALO
Main coordinator of the body.

**Spinal Cord**
MÉDULA ESPINAL
Nerve cable in vertebrae for reflexes.

**Reflex Action**
ACTO REFLEJO
Rapid involuntary defense movement.

**Voluntary Action**
ACTO VOLUNTARIO
Conscious decision made by cerebrum.

**Neuron**
NEURONA
Nerve cell transmitting electricity.

**Endangered**
EN PELIGRO DE EXTINCIÓN
Species at risk of disappearing.

**Habitat Loss**
PÉRDIDA DE HÁBITAT
Destruction of animals' natural homes.

**Poaching**
CAZA FURTIVA
Illegal hunting of protected animals.

**Faster than**
MÁS RÁPIDO QUE
Comparative form of fast.

**Stronger than**
MÁS FUERTE QUE
Comparative form of strong.

**Better than**
MEJOR QUE
Irregular comparative of good.

**More sensitive**
MÁS SENSIBLE
Comparative form of sensitive.

**Eyesight**
VISIÓN / VISTA
Ability to see through eyes.

**Reaction Time**
TIEMPO DE REACCIÓN
Time taken to answer a stimulus.

# 12. Speaking Arena • Diálogo de Clase

Práctica oral por parejas: una entrevista de campo entre dos jóvenes exploradores en Monfragüe

**SCOUT ALBA:** "Look up at the cliff! Is that a golden eagle soaring in the sky?"

**SCOUT HUGO:** "Yes! Its eyesight is **much sharper than** ours. It is spotting prey in the bushes."

**SCOUT ALBA:** "Listen to that sound! Are lynxes **more active than** wolves at twilight?"

**SCOUT HUGO:** "Yes! A lynx's hearing is **better than** ours, but its population is **more endangered**. We must protect its habitat!"

**SCOUT ALBA:** "Let's be quiet and observe through our binoculars without disturbing their territory."

# 13. CLIL Unit 2 Challenge!

4 retos interactivos para comprobar los conocimientos adquiridos en Ciencias e Inglés

## 👁️ Q1 • Senses

Which specific receptors on the retina detect light waves?

ANSWER:
**Photoreceptors! 💡**

## ⚡ Q2 • Reflexes

In an involuntary reflex action, where does the nerve loop turn around without waiting for the brain?

ANSWER:
**Spinal Cord! 🦴**

## 📝 Q3 • Grammar

Complete with comparative form: *"An eagle is _____ (fast) than a human runner."*

ANSWER:
**FASTER THAN! 🦅**

## 🐾 Q4 • Wildlife

Name the endangered Iberian feline protected in Extremadura with tufted ears for hearing:

ANSWER:
**The Iberian Lynx! 🐾**

# 🎉 1.ER TRIMESTRE COMPLETO!

¡Enhorabuena! Has dominado las dos grandes unidades del primer trimestre fusionando Ciencias e Inglés con éxito:

**⚡ Unit 1 Mastery:**
Células, orgánulos, niveles de organización, deportes de aventura, preposiciones y presente continuo.

**🦅 Unit 2 Mastery:**
5 sentidos, receptores, cerebro, actos reflejos, fauna amenazada (lince, ballena, águila) y comparativos.

